HKTM CCSDS Signal Generator

User Manual

CCSDS 131.0-B-5 test signal generation and recorded-pass analysis
for RF-Catcher (TestTree) Capture & Playback

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1 Introduction

1.1 Purpose

The **HKTM CCSDS Signal Generator** produces baseband IQ test signals that follow the CCSDS telemetry channel-coding chain (CCSDS 131.0-B-5) and writes them in a format that an IQ recorder/replayer can play back at RF – in particular the RF-Catcher (TestTree) Capture & Playback application. The signal can then be injected into a ground-station receiver to test demodulation, decoding, frame synchronisation and Virtual Channel routing with fully known content.

It also analyses real recorded passes (RF-Catcher `.rfcatcher` files or raw IQ): it measures symbol rate, carrier offset, Es/N0 and impairments, decodes the frames, and can copy the measured parameters back into the generator so that a recorded downlink can be reproduced.

1.2 What the tool provides

- **Signal generator**: Reed-Solomon, pseudo-randomizer, ASM, convolutional coding with puncturing, NRZ-L, QPSK/BPSK, RRC pulse shaping, all configurable.
- **Test features**: synthetic TM primary headers with selectable Spacecraft ID and Virtual Channels, deterministic Reed-Solomon error injection, transmitter impairments (LO offset, phase noise, IQ imbalance, PA compression) and channel Doppler.
- **Outputs**: raw interleaved IQ (float32 or int16) with a JSON metadata file, or a ready-to-play RF-Catcher `.rfcatcher` recording.
- **Recording analysis**: characterisation and decoding of real recorded passes.
- **Two interfaces**: a graphical interface (GUI) in the web browser with live spectrum and constellation, and command-line tools (CLI) for scripted use. Both use the same signal chain code.

1.3 Reference documents

Ref.	Document
[RD-1]	CCSDS 131.0-B-5, TM Synchronization and Channel Coding, Blue Book, September 2023
[RD-2]	CCSDS 132.0-B-3, TM Space Data Link Protocol, Blue Book (primary header only)
[RD-3]	ECSS-E-ST-50-01C, Space data links – Telemetry synchronization and channel coding
[RD-4]	TestTree RF-Catcher user manual (Capture & Playback application)
[RD-5]	PACKAGING.md and README.md in the software repository

1.4 Abbreviations

Term	Meaning
ASM	Attached Sync Marker (0x1ACFFC1D)
CADU	Channel Access Data Unit: ASM + Reed-Solomon codeblock(s)
Es/N0	Energy per symbol to noise power spectral density ratio
FER	Frame Error Rate
IQ	In-phase / Quadrature baseband samples
NRZ-L	Non-Return-to-Zero Level line coding
PA	Power Amplifier
PSD	Power Spectral Density
QPSK / BPSK	Quadrature / Binary Phase Shift Keying
RRC	Root-Raised-Cosine pulse-shaping filter (roll-off α)

Term	Meaning
RS	Reed-Solomon code
SCID / VCID	Spacecraft Identifier (10 bits) / Virtual Channel Identifier (3 bits)

2 Installation and start-up

The tool can be used in two ways: as a **standalone Windows executable** (no Python needed), or **from the source code** with Python. The functions are the same.

2.1 Standalone executable

What you receive

A zipped folder `HKTM-CCSDS-Signal-Generator` and, separately, a licence file `license.key`.

Installation

- Unzip the folder anywhere (for example `C:\Tools\`). No installer and no administrator rights are needed.
- Copy `license.key` into that folder, next to `HKTM-CCSDS-Signal-Generator.exe`.
- Keep the `_internal` folder next to the `.exe`: it contains Python and the libraries. The folder is about 500 MB.

```
HKTM-CCSDS-Signal-Generator\
    HKTM-CCSDS-Signal-Generator.exe
    license.key            <- the licence file you received
    _internal\             <- do not move or delete
```

Starting the program

- Double-click `HKTM-CCSDS-Signal-Generator.exe`.
- A splash screen with the logo shows the start-up progress, and a console window opens.
- After a few seconds the GUI opens in the default web browser at `http://localhost:8501`. If the browser does not open, type that address by hand (the console window also prints it).
- The program runs only on your computer: the browser is just its display. Nothing is sent over the network.
- **To stop the program, close the console window.** Closing the browser tab alone leaves it running.

Licence

The executable checks the licence file at every start. It holds the customer name and the last valid day, and is signed by the issuer: any change to the file makes it invalid. The sidebar of the GUI shows Licensed to <customer> · valid until <date>; during the last 30 days a warning is shown instead.

Situation	What happens	What to do
No `license.key` in the folder	The console prints “No licence found” with the folder to use, and the program stops.	Copy the licence file next to the `.exe`.
Licence expired	The console prints the expiry date and the program stops.	Ask the issuer for a new licence file (no new executable is needed).
File altered or damaged	“The licence file is not valid” / “damaged”.	Use the original file as received.
System date set back	“The system date appears to have been set back”.	Correct the date and time of the computer and restart.
Expires within 30 days	The program works; the GUI shows a warning.	Request a renewal.

Table 1 – Licence messages at start-up

2.2 Running from source

Requires Python 3.10 or later. In a terminal, in the repository folder:

```
pip install -r requirements.txt
streamlit run app.py            # GUI, opens in the browser
python generate_signal.py -h    # CLI help
```

When run from source there is no licence check.

3 The signal chain

The generator applies the CCSDS channel-coding steps in the order of [RD-1]. Each step can be configured or switched off:

```
payload -> RS(255,223) interleave x5 -> [pseudo-randomizer, ASM excluded]
        -> + ASM (per CADU)  -> convolutional K=7 r=1/2 (over the CADU stream)
        -> NRZ-L -> QPSK (Gray) or BPSK -> RRC or no filter -> [impairments] -> [Doppler]
        -> normalisation -> [resampling] -> IQ file
```

3.1 Baseline parameters

Parameter	Baseline value	Options
Modulation	QPSK (Gray)	QPSK, BPSK
Symbol rate	1 785 000 S/s	any; or set the bit rate (bit rate = symbol rate × 2 for QPSK)
Bit rate (channel)	3 570 000 bps	derived
Line coding	NRZ-L (1 → +1, 0 → −1)	NRZ-L only
Reed-Solomon	RS(255,223), E = 16, interleave depth I = 5	on/off; E = 8 or 16; I = 1, 2, 3, 4, 5, 8
Pseudo-randomizer	CLI: none · GUI: long	none; short (255-bit, legacy); long (131071-bit, CCSDS default since 2023)
ASM	0x1ACFFC1D, 4 bytes, not coded by RS	fixed
CADU length	4 + 255 × I = 1279 bytes	depends on I
Convolutional	K = 7, rate 1/2, G1 = 171o, G2 = 133o, G2 inverted	on/off; rate 1/2, 2/3, 3/4, 5/6, 7/8
Pulse shaping	RRC, α = 0.35, span 8 symbols, 4 samples/symbol	RRC (α, span) or none (rectangular NRZ pulses); samples/symbol
Spacecraft ID	0x123 (291)	0–1023; written only with Virtual Channels enabled
Output	float32 IQ, peak 0.9, native rate (7.14 MS/s)	int16, .rfcatcher, peak amplitude, resampling to a fixed rate

Table 2 – Baseline parameters

3.2 Input formats

The payload can be pseudo-random (reproducible with a seed) or read from a file. The input format says what the data already contains, so that no step is applied twice:

Input format	File content	Steps applied by the generator
Transfer Frame (default)	Raw, uncoded Transfer Frames	RS, randomizer, ASM, convolutional, modulation
ASM + Transfer Frame	ASM + frame, not scrambled (e.g. decoded output of a ground station, or frames saved by the recording analysis)	Randomizer (never on the ASM; default short, “none” not allowed), convolutional, modulation. No RS, no new ASM.
CADU as on air	ASM + already RS-coded and scrambled block	Convolutional and modulation only (randomizer must be “none”)

Table 3 – Input formats

With file input, each record is exactly 4 + 255 × I bytes. The tool searches for the first real ASM, so a file need not start on a CADU boundary, and skips any recorder-added bytes between the end of one CADU and the next ASM.

4 Graphical interface (GUI)

The GUI has a sidebar with the generator settings and two tabs: **Signal generator** and **Recording analysis**. Every change of a setting recomputes a short preview of the signal immediately.

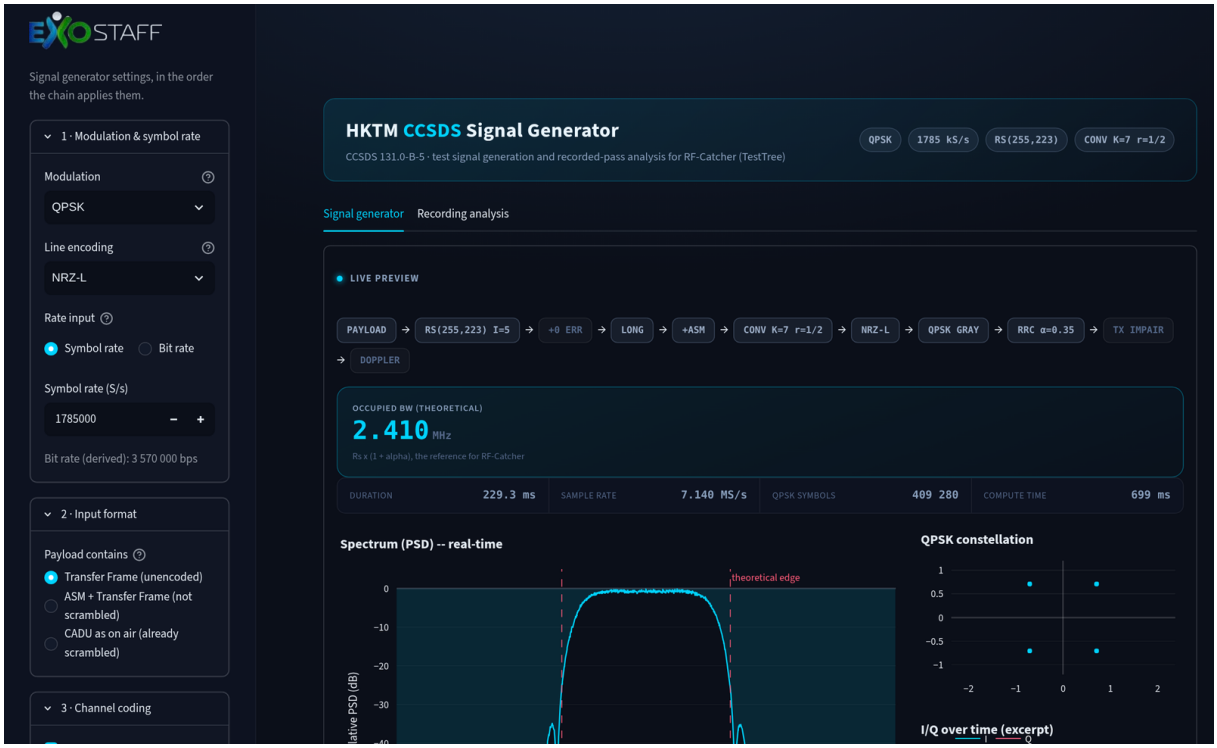


Figure 1 – Main window: settings sidebar, header with the active baseline, the two tabs and the live preview of the signal generator

4.1 Settings sidebar

The settings are grouped in the order the chain applies them. Each field has a help icon (?) with a short explanation.

Section	Content
1 · Modulation & symbol rate	QPSK/BPSK, line coding, symbol rate or bit rate (the other is derived).
2 · Input format	What the payload contains (Table 3).
3 · Channel coding	Reed-Solomon on/off, E, interleave depth; convolutional on/off, code rate, G2 inversion; pseudo-randomizer.
4 · Payload & duration	Pseudo-random (seed) or file; preview CADU count; **Assign Virtual Channels** (VCID list) and **Spacecraft ID**.
5 · Deterministic error injection	Exact number of RS symbol errors in a chosen codeword of chosen CADUs or of one Virtual Channel.
6 · Pulse shaping	RRC filter (roll-off α, span) or **None (NRZ, rectangular)**; samples per symbol; optional **Butterworth output filter** (order, −3 dB bandwidth).
7 · Impairments & Doppler	LO frequency offset and phase noise; IQ gain and phase imbalance; PA saturation and AM-PM; channel Doppler shift and rate.

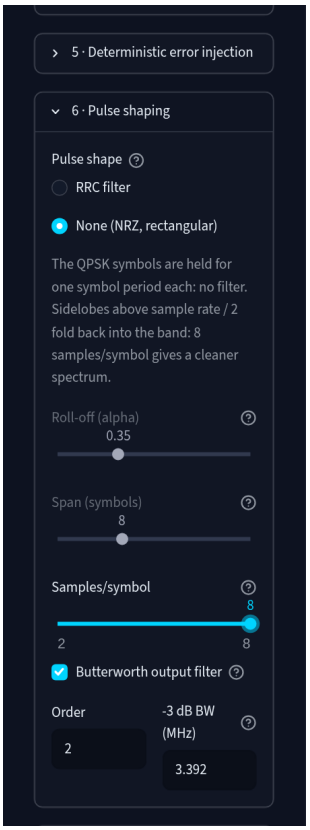


Figure 2 – Sidebar sections (right: section 6 with no pulse shaping and the Butterworth output filter)

4.2 Signal generator tab: live preview

The **Live preview** panel shows:

- the chain stages as a row of labels: active stages are highlighted, disabled ones are greyed out;
- the theoretical occupied bandwidth (RRC: Rs × (1 + α); no filter: 2 × Rs), duration, sample rate, number of symbols and compute time of the preview;
- the **spectrum** (PSD) with the theoretical band edges as dashed red lines;
- the **constellation**, taken after a matched filter, so impairments (phase noise, IQ imbalance, PA) are visible;
- an **I/Q excerpt** over time.

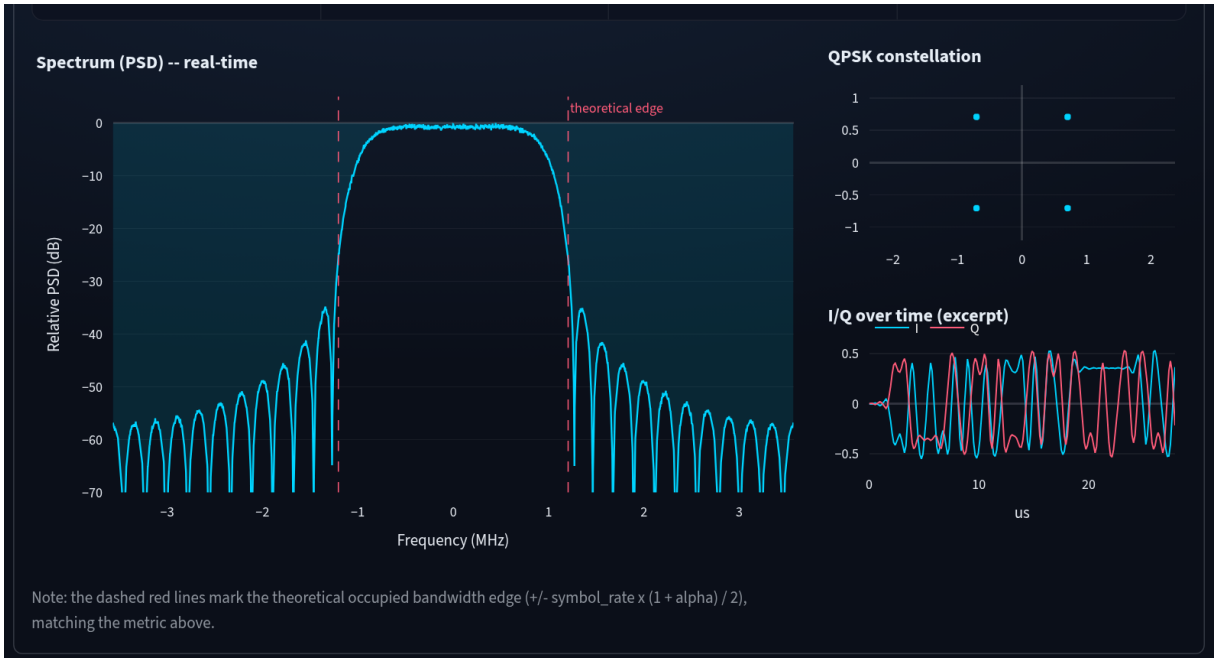


Figure 3 – Live preview: spectrum with theoretical band edges, constellation and I/Q excerpt

The preview uses only a few tens of CADUs to stay fast. The length of the exported file is set separately in the Export panel.

4.3 Signal generator tab: export

The **Export** panel writes the full-length file:

Field	Meaning
Export CADU count	Length of the file. The panel shows the resulting duration and file size.
Output format	**RF-Catcher recording (.rfcatcher)**: carries its own sample rate, RF frequency and bandwidth. **Raw IQ int16** (RF-Catcher sample format) or **raw IQ float32**: no header; rate and frequency must be set by hand on the replayer.
Peak amplitude	Normalised peak (default 0.9), headroom against clipping.
Resample to fixed rate	Resample to an exact rate accepted by the replayer, e.g. 10 000 000 Hz. Without it, the native rate is symbol rate × 4.
RF center frequency	Written into the .rfcatcher metadata (default 1707 MHz).
Bandwidth: automatic	RF-Catcher's linked value, sample rate / 1.1; untick to enter it.
Device metadata from	Optional real .rfcatcher recording whose device fields (serial number, firmware, gain) are copied.
Generate export file	Creates the file in the `output` folder and offers the download buttons (file and metadata). Files above 1 GB stay in the folder only.

Table 4 – Export panel

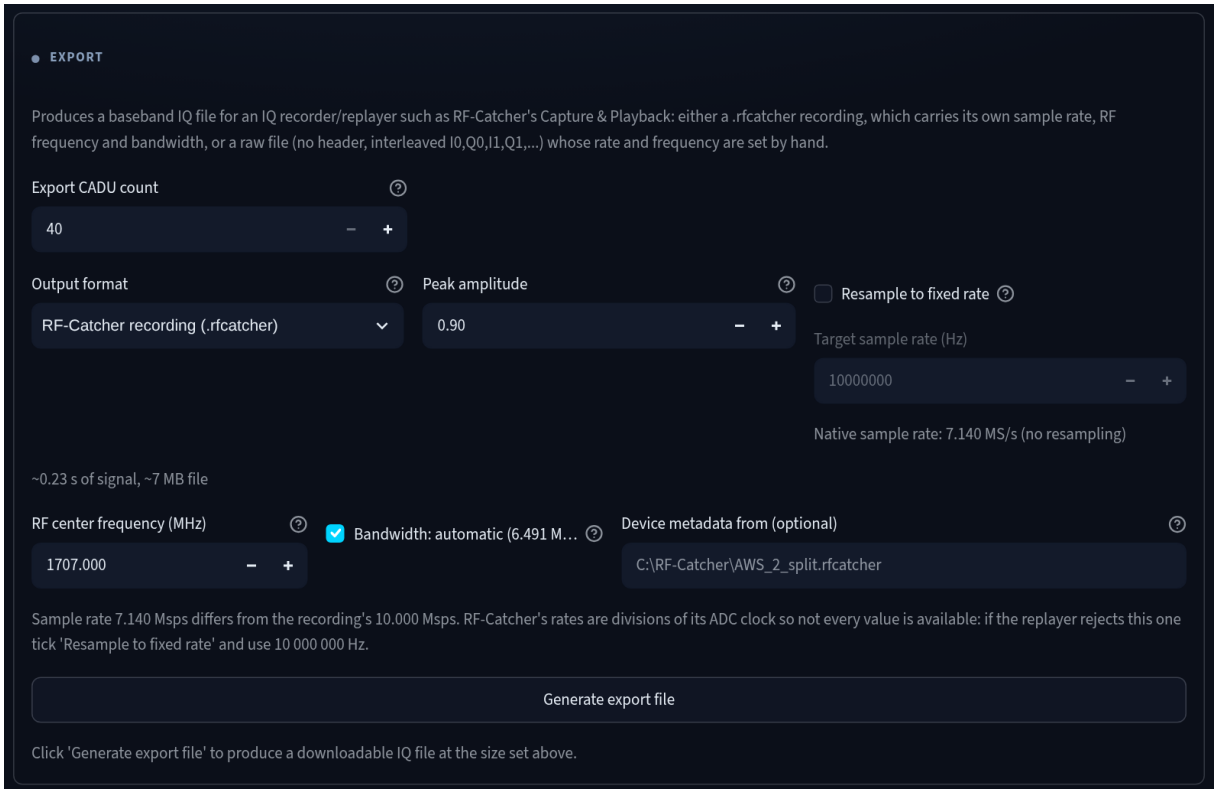


Figure 4 – Export panel with the .rfcatcher output format

RF-Catcher only accepts some sample rates (divisions of its ADC clock). If the replayer refuses the native rate, tick **Resample to fixed rate** and use 10 000 000 Hz. The panel warns when the rate differs from the template recording.

4.4 Recording analysis tab

This tab characterises a real recorded pass and can transfer the result to the generator.

- Enter the **local path** of the recording (.rfcatcher or raw IQ). The file is read directly from disk, not uploaded, so multi-GB files are fine.
- Choose the slice: **Start at** and **Duration** (up to 2 s; 1 s is plenty). Only that slice is read.
- For raw IQ, give the sample rate and sample format; for .rfcatcher they come from the metadata.
- Nominal symbol rate and matched-filter roll-off are starting values for the estimation.
- **Decode frames** runs the Viterbi decoder and the frame analysis (slower: about 15 s).
- Click **Analyze recording**.

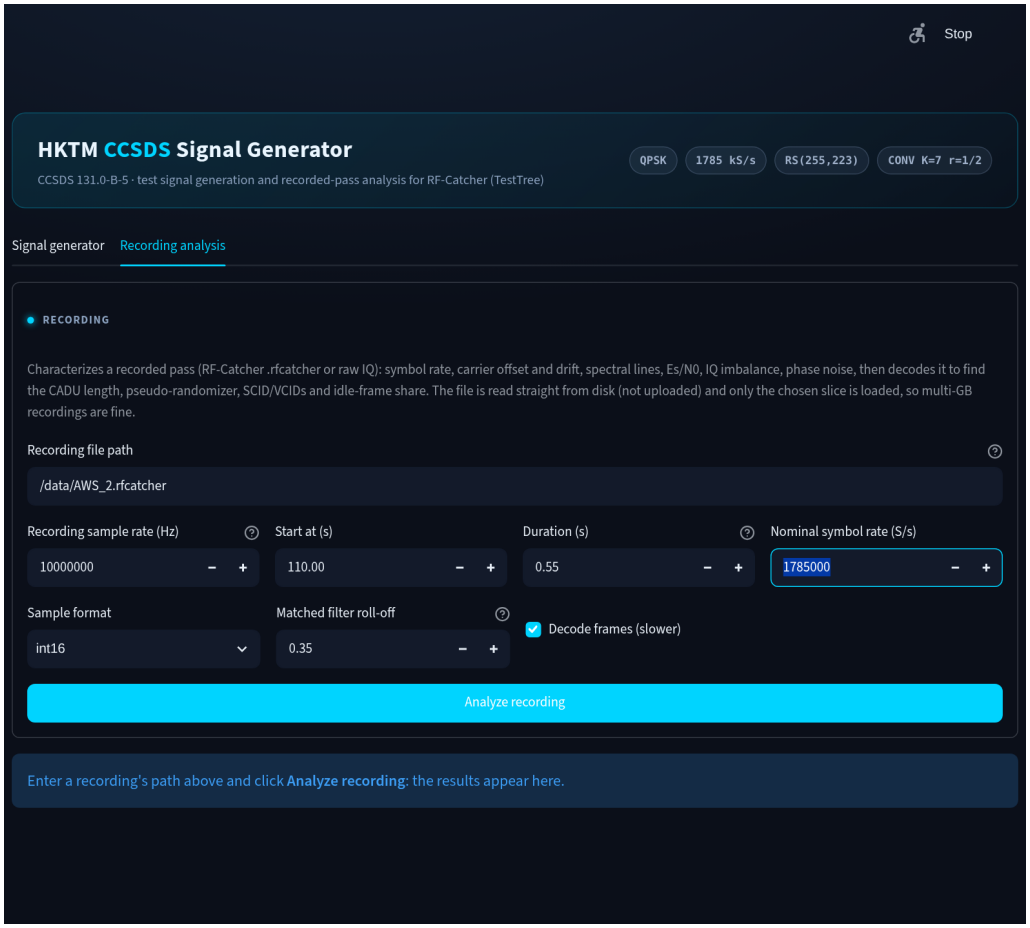


Figure 5 – Recording analysis: input fields

The results show symbol rate (and offset in ppm), carrier offset and drift, Es/N0 and EVM, the number of discrete spectral lines, the randomizer and CADU length found, the share of idle frames and the Spacecraft ID; then the spectrum, the constellation after carrier recovery and the residual carrier phase. **Full report** lists all values, including IQ imbalance, phase noise and the Virtual Channels found.



Figure 6 – Recording analysis: results and the “Reproduce it in the generator” panel

Reproduce it in the generator

Tick the values to copy (symbol rate, rounded to 50 Hz; modulation; randomizer; optionally the carrier offset) and click **Apply to generator**: the generator tab opens with those settings. **Download the decoded frames** saves them as ASM + Transfer Frame records; load that file as payload with input format “ASM + Transfer Frame” to regenerate the recorded content bit for bit.

5 Command-line tools (CLI)

Three scripts cover the same functions as the GUI. All options are listed in Appendix A and by `-h`.

5.1 generate_signal.py – generate a signal

```
python generate_signal.py                       # baseline, file in output/
python generate_signal.py --n-cadu 500 --randomizer long -o test.raw
python generate_signal.py --rs-e 8 --interleave-depth 2 --conv-rate 3/4 -o test2.raw
python generate_signal.py --modulation BPSK -o test3.raw
python generate_signal.py --dtype int16 --target-fs 10e6 --peak 0.9
python generate_signal.py --rfcatcher --target-fs 10e6 --rf-frequency-mhz 1707
```

Without `-o`, the file is written to `output/qpsk_ccsds_<fs>Msps_<n_cadu>cadu_<timestamp>.iq`. Generation runs in batches, so memory use does not grow with the file length (resampling is the only exception: very long files with resampling are refused before starting).

5.2 analyze_recording.py – analyse a recording

```
python analyze_recording.py C:\RF-Catcher\AWS_2_split.rfcatcher --offset 30 --duration 1 --plot aws2.png
python analyze_recording.py capture.iq --fs 10e6 --dtype int16 --save-frames frames.bin
```

It prints the same report as the GUI. `--plot` saves spectrum, constellation and residual phase as PNG; `--save-frames` writes the decoded frames in the "ASM + Transfer Frame" input format. About 15 s and 2 GB of memory per second of signal, plus about 15 s of decoding (limited by default to the first 400 000 symbols, about 40 CADUs).

5.3 verify_spectrum.py – check an exported file

```
python verify_spectrum.py output/qpsk_ccsds_....iq --plot spectrum.png
```

Computes the PSD of a generated file (reading its .meta.json for rate and format) and measures the −3 dB and null-to-null bandwidths. For the RRC baseline (Rs = 1.785 MS/s, α = 0.35): −3 dB ≈ Rs, null-to-null ≈ Rs (1 + α) ≈ 2.41 MHz. Without filter (NRZ): −3 dB ≈ 0.886 × Rs, null-to-null = 2 × Rs ≈ 3.57 MHz.

The −3 dB edge is measured on the PSD smoothed over 31 frequency bins and referred to its smoothed peak, so the result is reliable also for short files (a few CADUs), whose raw PSD estimate scatters by a few dB from bin to bin. The null-to-null width is measured on the raw PSD.

6 Test scenarios

Typical uses, with the CLI command. The same settings are available in the GUI sidebar.

6.1 Virtual Channels and Spacecraft ID

Builds a real 6-byte TM primary header [RD-2] into each frame: Spacecraft ID, Virtual Channel ID (round-robin over the list), master and virtual channel frame counters. Repeating a VCID gives it a larger share. Use it to check that a receiver identifies and routes frames per Virtual Channel.

```
python generate_signal.py --vcid-list 0,0,1,2 --spacecraft-id 291 -o vc_test.raw
```

The Spacecraft ID (10 bits, 0–1023, default 0x123 = 291) is written only together with Virtual Channels (`--vcid-list` / "Assign Virtual Channels"). It cannot be combined with a payload file, because it would overwrite the first 6 bytes of real data.

6.2 Reed-Solomon boundary test

Flips an exact number of RS symbols in one codeword of chosen CADUs. With E = 16, 16 errors must be corrected; 17 must be detected as uncorrectable and never miscorrected.

```
python generate_signal.py --corrupt-rs-symbols 16 --corrupt-cadu-indices 10 -o rs16.raw
python generate_signal.py --corrupt-rs-symbols 17 --corrupt-cadu-indices 10 -o rs17.raw
```

6.3 Per-Virtual-Channel error accounting

```
python generate_signal.py --vcid-list 0,1,2 --corrupt-rs-symbols 17 --corrupt-vc 1 -o vc1_err.raw
```

Every VC 1 frame is uncorrectable; the receiver's FER report must show errors on VC 1 only.

6.4 Transmitter impairments

```
python generate_signal.py --freq-offset-hz 5000 --iq-gain-imbalance-db 0.8 \
    --iq-phase-imbalance-deg 3 --phase-noise-linewidth-hz 200 -o tx_impair.raw
python generate_signal.py --pa-backoff-db -6 --pa-smoothness 3 --pa-am-pm-deg-per-db 3 -o pa_sat.raw
```

Checks carrier recovery, image rejection and tolerance to EVM. A PA driven into saturation produces spectral regrowth outside the occupied band.

6.5 Unfiltered NRZ signal

Some satellites (AWS, for example) transmit QPSK without pulse shaping: rectangular NRZ pulses, followed at most by an output band-pass filter. Select **None (NRZ, rectangular)** in sidebar

section 6, or on the CLI:

```
python generate_signal.py --pulse-shape nrz --sps 8 -o nrz.raw
```

The spectrum is a sinc²: first nulls at ±Rs (null-to-null 2 × Rs = 3.57 MHz, the same main lobe as an RRC with α = 1) and sidelobes at about −13, −18, −21 dB that decay slowly and are not band-limited. Use 8 or more samples per symbol: at 4, the sidelobes above half the sample rate fold back into the band. Resampling to a lower rate removes them above the new Nyquist frequency. The constellation is computed with an integrate-and-dump (rectangular) matched filter.

Output filter

A real transmitter without pulse shaping still has an output filter after the power amplifier, which lowers the sidelobes. Tick **Butterworth output filter** in section 6 (order and −3 dB bandwidth), or on the CLI:

```
python generate_signal.py --pulse-shape nrz --sps 8 --output-filter-order 2 -o aws_like.raw
```

The filter is an RF band-pass centred on the carrier, −3 dB bandwidth 1.9 × Rs by default (`--output-filter-bw-hz` to change it). With order 2 the sidelobes drop to about −22, −35, −42 dB, close to the −21, −34, −45 dB measured on AWS. It reproduces the analog filter's response within about 1 dB up to 0.45 × the sample rate. Like a real filter it delays the signal and adds a little inter-symbol interference: the constellation is shown after removing the delay, as a receiver's timing recovery would, so only the interference remains (small spread of each point).



Figure 7 – Live preview with no pulse shaping and a 2nd-order Butterworth output filter: sinc² main lobe with lowered sidelobes, first nulls marked

6.6 Doppler of a pass

```
python generate_signal.py --doppler-hz 30000 --doppler-rate-hz-s -400 -o doppler.raw
```

Frequency shift at the start of the file plus a constant rate (Hz/s): a linear approximation of the Doppler curve of a LEO pass, to check the receiver's carrier tracking.

6.7 Reproducing a recorded pass

- Analyse 1 s of the recording (GUI Recording analysis tab, or `analyze_recording.py --save-frames frames.bin`).
- Apply symbol rate, modulation and randomizer to the generator.
- Load the decoded frames as payload with input format “ASM + Transfer Frame”.
- Export as .rfcatcher at the recorder's sample rate and replay it.

7 Output files

File	Content
`*.iq` / `*.raw` (float32)	Interleaved I0, Q0, I1, Q1, … as 32-bit floats in [−1, +1]. No header.
`*.iq` / `*.raw` (int16)	Interleaved 16-bit little-endian integers, 12 significant bits, range [−2048, 2047] (RF-Catcher format). No header.
`*.meta.json`	All parameters used, actual sample rate and format of the file. Note: a raw file is baseband; the RF frequency must be set on the replayer.
`*.rfcatcher`	RF-Catcher recording: a tar archive of `<name>.iq` (int16) and `<name>.json` (sample rate, RF frequency, bandwidth, duration, device fields). Headers are written exactly as RF-Catcher writes them.

Table 5 – Output files

All files are written to the `output` folder unless another path is given.

8 Licence administration (issuer)

This chapter is for the person who distributes the executable. Users only need section 2.1. The tool `license_tool.py` is run from the source code and is not part of the executable.

Step	Command	Notes
1. Once: key pair	`python license_tool.py keygen --private-key C:\secure\hktm_license_private_key.pem`	The private key stays with the issuer, outside the repository. The public key is written to assets/license_public_key.pem and is built into every executable made from then on.
2. Build	`pyinstaller hktm_simulator.spec`	See PACKAGING.md. Builds without the public key have no licence check.
3. Per customer	`python license_tool.py issue --private-key C:\secure\hktm_license_private_key.pem --customer "Name" --expires 2027-12-31 -o license.key`	The expiry date is the last valid day. Send license.key with the zipped folder.
4. Renewal	`Same as step 3 with a new date`	No new executable needed.
Check	`python license_tool.py verify license.key`	Prints VALID or the reason.

Table 6 – Licence administration

The licence is a deterrent, not copy protection. Never commit the private key or issued licence files; both are excluded by .gitignore. If the private key is lost, a new key pair, new licences and a rebuild are needed.

Appendix A CLI option reference

Generated from the scripts' own argument parsers when this manual was built.

A.1 generate_signal.py

Option	Default	Description
`--n-cadu N_CADU`	`100`	See section 5.
`--modulation {QPSK,BPSK}`	`QPSK`	QPSK (baseline, 2 bits/symbol) or BPSK (1 bit/symbol, half the bit rate at the same symbol rate -- the modulation the base rate-1/2 convolutional code's G2 inversion, CCSDS 3.3.1(5), is specified against)
`--pulse-shape {rrc,nrz}`	`rrc`	'rrc': Root-Raised-Cosine filter (--alpha, --span); 'nrz': no filter, rectangular pulses one symbol long (sinc^2 spectrum, first nulls at +/-symbol rate, sidelobes decaying as 1/f^2), like a transmitter without pulse shaping. With nrz the sidelobes above sample rate/2 fold back: use --sps 8 or more for a cleaner spectrum
`--alpha ALPHA`	`0.35`	RRC roll-off (rrc only)
`--sps SPS`	`4`	samples per symbol (native oversampling)
`--span SPAN`	`8`	RRC length in symbols (rrc only)
`--no-rs`	`off`	disable Reed-Solomon FEC
`--rs-e {8,16}`	`16`	RS error correction capability
`--interleave-depth {1,2,3,4,5,8}`	`5`	See section 5.
`--no-conv`	`off`	disable convolutional FEC
`--conv-rate {1/2,2/3,3/4,5/6,7/8}`	`1/2`	See section 5.
`--no-invert-g2`	`off`	See section 5.
`--randomizer {none,short,long}`	—	CCSDS pseudo-randomizer: 'long' (131071-bit, current standard default), 'short' (255-bit, legacy), or 'none'. Default: 'short' with --input-format asm_frame (where 'none' is rejected), otherwise 'none'. Must be 'none' with --input-format cadu
`--input-format {transfer_frame,asm_frame,cadu}`	`transfer_frame`	what --payload-source/the generated payload represents: 'transfer_frame' (raw, uncoded data -- RS/randomizer/ASM applied here, default), 'asm_frame' (ASM + Transfer Frame, NOT scrambled -- scrambled here, never the ASM) or 'cadu' (CADU exactly as on the air, already scrambled -- used verbatim). Neither ASM-framed format gets RS or a new ASM; each record is 4 + 255*I bytes, extra bytes before the next ASM are skipped. The convolutional stage still runs on all formats
`--payload-source PAYLOAD_SOURCE`	—	See section 5.
`--seed SEED`	`42`	See section 5.
`--vcid-list VCID_LIST`	—	comma-separated Virtual Channel IDs (0-7), e.g. '0,1,2', round-robined across generated Transfer Frames: builds a real CCSDS TM primary header (132.0-B-3 4.1.2) carrying each frame's assigned VCID, for validating a receiver correctly identifies/routes frames by Virtual Channel. Repeat a value to weight it more heavily (e.g. '0,0,1' gives VC 0 twice VC 1's share). Only valid with --input-format transfer_frame and no --payload-source (it would overwrite the first 6 bytes of real data); default: no synthetic header, unchanged payload
`--spacecraft-id SPACECRAFT_ID`	`291`	10-bit Spacecraft ID (0-1023) for the synthetic primary header when --vcid-list is set; otherwise unused

Option	Default	Description
`--corrupt-rs-symbols CORRUPT_RS_SYMBOLS`	0	deterministically flip exactly this many RS symbols within one interleaved codeword (--corrupt-codeword-index) of each targeted CADU (--corrupt-cadu-indices and/or --corrupt-vc), to boundary-test a receiver's RS decoder against its declared correction capability E (--rs-e): E errors must still decode perfectly, E+1 must fail -- an exact, reproducible per-codeword error count a replayer's AWGN can't guarantee hitting. Needs --no-rs NOT set (or --input-format cadu, already RS-coded); default: 0 (disabled)
`--corrupt-codeword-index CORRUPT_CODEWORD_INDEX`	0	which of the --interleave-depth interleaved RS codewords to target (0-based); only used with --corrupt-rs-symbols
`--corrupt-cadu-indices CORRUPT_CADU_INDICES`	–	comma-separated 0-based CADU indices to corrupt, e.g. '0,5,10'; combines with --corrupt-vc (either match corrupts that CADU) -- at least one of the two is required with --corrupt-rs-symbols
`--corrupt-vc CORRUPT_VC`	–	corrupt every CADU assigned to this Virtual Channel ID (requires --vcid-list) -- e.g. to verify a receiver's per-VC FER accounting attributes errors to the right channel only
`--corrupt-seed CORRUPT_SEED`	777	seed for which symbol positions/values get flipped -- deterministic and reproducible across runs
`--freq-offset-hz FREQ_OFFSET_HZ`	0.0	residual LO frequency offset applied to the pulse-shaped IQ, in Hz (positive or negative) -- for validating a receiver's carrier-recovery loop against a real (imperfect) transmitter LO, not just a perfectly on-frequency signal; default: 0 (disabled)
`--iq-gain-imbalance-db IQ_GAIN_IMBALANCE_DB`	0.0	IQ modulator I/Q branch gain mismatch, in dB -- produces a mirror-image tone in the spectrum; default: 0 (disabled)
`--iq-phase-imbalance-deg IQ_PHASE_IMBALANCE_DEG`	0.0	IQ modulator phase deviation from ideal 90-degree I/Q separation, in degrees -- combines with --iq-gain-imbalance-db in the same mirror-image model; default: 0 (disabled)
`--phase-noise-linewidth-hz PHASE_NOISE_LINEWIDTH_HZ`	0.0	free-running-oscillator single-sideband 3 dB linewidth, in Hz -- generates Wiener (random-walk) phase noise on the IQ, for validating a receiver's tolerance to constellation smearing from a real transmitter LO; default: 0 (disabled)
`--impairment-seed IMPAIRMENT_SEED`	2718	seed for the phase noise random walk -- deterministic and reproducible across runs
`--doppler-hz DOPPLER_HZ`	0.0	channel Doppler shift, in Hz, at the start of the exported signal (not a transmitter impairment -- applied last, after the transmitter chain, since it's imparted by relative spacecraft/ground-station motion along the propagation path); positive or negative; default: 0 (disabled)
`--doppler-rate-hz-s DOPPLER_RATE_HZ_S`	0.0	constant Doppler rate of change, in Hz/s -- combined with --doppler-hz as a local-linear (chirp) approximation of a real LEO pass's Doppler curve, for validating a receiver's carrier-tracking loop against realistic pass dynamics, not just a static frequency error; default: 0 (disabled)
`--pa-backoff-db PA_BACKOFF_DB`	–	power amplifier saturation point (Rapp AM-AM model), in dB above this project's reference unit amplitude (1.0, an ideal symbol's own magnitude) -- models PA compression/saturation, the last physical stage before the antenna: negative values saturate more aggressively (visible spectral regrowth just outside the occupied bandwidth); default: not set (disabled)
`--pa-smoothness PA_SMOOTHNESS`	3.0	Rapp model knee sharpness (higher = sharper transition from linear to saturated); only used with --pa-backoff-db
`--pa-am-pm-deg-per-db PA_AM_PM_DEG_PER_DB`	0.0	AM-PM conversion, in degrees of phase shift per dB of AM-AM compression (the same figure real TWTA/SSPA datasheets quote); 0 (default) disables it; only used with --pa-backoff-db

Option	Default	Description
`--output-filter-order OUTPUT_FILTER_ORDER`	`0`	Butterworth transmitter output filter after the PA, order 1-10 (0 = none, default). A low-order filter on an unfiltered NRZ signal (--pulse-shape nrz) lowers its sidelobes the way a real transmitter's output filter does: order 2 with the default bandwidth gives about -22/-35/-42 dB, close to AWS
`--output-filter-bw-hz OUTPUT_FILTER_BW_HZ`	–	RF -3 dB bandwidth of the output filter, two-sided, in Hz; default 1.9 x symbol rate; only used with --output-filter-order
`--dtype {float32,int16}`	`float32`	IQ sample format for the output file; int16 is the RF-Catcher format (little-endian, 12 significant bits, range [-2048, 2047])
`--peak PEAK`	`0.9`	normalized peak amplitude before dtype conversion (headroom against clipping)
`--target-fs TARGET_FS`	–	resample the output to this exact sample rate in Hz (e.g. 10e6); default: no resampling, native symbol_rate*sps
`--rfcatcher`	`off`	write an RF-Catcher recording (.rfcatcher: int16 IQ + JSON metadata with sample rate, RF frequency and bandwidth) instead of a raw IQ file; implies --dtype int16
`--rf-frequency-mhz RF_FREQUENCY_MHZ`	`1707.0`	RF center frequency written to the .rfcatcher metadata (default 1707 MHz)
`--rf-bandwidth-mhz RF_BANDWIDTH_MHZ`	–	bandwidth written to the .rfcatcher metadata (default: RF-Catcher's linked value, sample rate / 1.1)
`--rfcatcher-template RFCATCHER_TEMPLATE`	–	a real .rfcatcher recording (or its .json) whose device fields (serial, firmware, gain, level) are copied into the new file's metadata; default: built-in AWS_2 ones
`-o, --output OUTPUT`	–	output path; default: output/qpsk_ccsds_<fs>Msps_<n_cadu>cadu_<timestamp>.iq

A.2 analyze_recording.py

Option	Default	Description
`recording`	`(required)`	.rfcatcher (tar) or raw interleaved IQ file
`--fs FS`	–	recording sample rate, Hz (e.g. 10e6); default: read from the .rfcatcher metadata
`--dtype {int16,float32}`	`int16`	See section 5.
`--header-bytes HEADER_BYTES`	`0`	bytes to skip in a raw (non-tar) file
`--offset OFFSET`	`0.0`	start of the analyzed slice, s
`--duration DURATION`	`1.0`	length of the analyzed slice, s
`--rs-nominal RS_NOMINAL`	`1785000.0`	expected symbol rate, sps
`--alpha ALPHA`	`0.35`	RRC roll-off of the matched filter
`--max-carrier-offset MAX_CARRIER_OFFSET`	`200000.0`	carrier search range, +-Hz
`--modulation {auto,QPSK,BPSK}`	`auto`	See section 5.
`--decode-symbols DECODE_SYMBOLS`	`400000`	max symbols to Viterbi-decode (default ~40 CADUs, ~15 s; roughly 20-30 s per million)
`--no-decode`	`off`	See section 5.
`--plot PLOT`	–	save a spectrum/constellation/phase PNG here
`--save-frames SAVE_FRAMES`	–	write the decoded frames here as ASM + de-randomized Transfer Frame records (the generator's 'asm_frame' input format)

A.3 verify_spectrum.py

Option	Default	Description
`iq_file`	`(required)`	See section 5.
`--sample-rate SAMPLE_RATE`	–	override meta.json output_sample_rate (Hz)
`--dtype {float32,int16}`	–	override meta.json output_dtype
`--nperseg NPERSEG`	`4096`	See section 5.
`--plot PLOT`	–	save spectrum plot PNG to this path